

nyc-college-mission-vision-statements

March 16, 2026

1 Project Two Executive Summary

1.1 Mission and Vision Statement Networks in New York City Colleges

This study examines how colleges in New York City frame their mission and vision statements.

Using publicly available college website data, and data from the [Integrated Postsecondary Education Data System \(IPEDS\)](#), we constructed a dataset of NYC college mission and vision statements. The statements were copied and pasted into a Google Sheet and then an API is used to connect the data to the Jupyter Notebook.

1.1.1 Model 1: Preparing the Tools

In this section, I loaded the software libraries needed to complete my analysis.

In sum, these tools allow me to:

- Create a dictionary of mission and vision statements for colleges.
- Perform calculations and statistical tests.
- Create visual charts to help interpret results.

Even though some of these libraries are basic and repeatedly used in many projects, such as pandas, I commented out each library to understand their functions.

```
[1]: # Module 1: Import Libraries

import pandas as pd                # Used to work with tables of data (like
    ↪Excel sheets in Python)

from google.oauth2.service_account import Credentials # Used to authenticate
    ↪and connect to Google Sheets API

import gspread                    # Used to interact with Google Sheets
    ↪through the API

# import matplotlib.pyplot as plt    # Used to control chart formatting and
    ↪display plots

from sklearn.feature_extraction.text import TfidfVectorizer # Used to convert
    ↪text into numerical features using TF-IDF
```

```
import re # Used to perform regular expression
    ↪ matching and text manipulation

import networkx as nx # Used to create and analyze networks
    ↪ and graphs
```

1.1.2 Module 2: Loading the Data

From Google Sheets, we set up an API to directly read in the data.

In addition to the mission and vision statements, some of the variables contained in this dataset include, but are not limited to:

1. UNITID (A unique identifier for each institution).
2. Institutional Name.
3. Address info included longitude and latitude geodata to potential map institutions.

```
[2]: import gspread
import pandas as pd
from google.oauth2.service_account import Credentials

scope = [
    "https://www.googleapis.com/auth/spreadsheets",
    "https://www.googleapis.com/auth/drive"
]

creds = Credentials.from_service_account_file(
    r"C:\Users\zengo\OneDrive\Magic Briefcase\School of Professional
    ↪ Studies\DATA 620 Web Analytics\Project 2\crucial-zodiac-333018-7db0dfea33da.
    ↪ json",
    scopes=scope
)

client = gspread.authorize(creds)

# ONLY the spreadsheet ID
spreadsheet = client.open_by_key("1nE5HAiaJ1hnqT8L-6V39qtfZdmgrvhg0xJ8USGvzLrk")

worksheet = spreadsheet.get_worksheet(0)

data = worksheet.get_all_records()

college_statements = pd.DataFrame(data)

college_statements.head()
```

```

[2]:  UNITID                                INSTNM \
0  188340      Vaughn College of Aeronautics and Technology
1  188650                                Allen School-Brooklyn
2  188678      American Academy of Dramatic Arts-New York
3  188687  American Academy McAllister Institute of Funer...
4  188890                                Apex Technical School

      CITY STABBR    ZIP  LATITUDE  LONGITUD \
0      Flushing    NY  11369  40.768348 -73.881721
1      Brooklyn    NY  11201  40.693797 -73.991546
2      New York    NY  10016  40.745545 -73.984843
3      New York    NY  10036  40.757105 -73.986466
4  Long Island City    NY  11101  40.748807 -73.949306

                                Mission \
0  To provide a dynamic learning environment buil...
1  To provide superior, high-quality, campus-base...
2  A world-leading acting conservatory and colleg...
3  AAMI exists to: Provide a learning environment...
4  Offer quality hands-on training that provides ...

                                Vision
0  To change the world one student at a time with...
1  Allen College is a Leading Institution to Enri...
2  The American Academy of Dramatic Arts upholds ...
3
4
                                None
                                None

```

1.1.3 Data Diagnostics

In Modules 3A-I, we apply various functions to diagnose the data.

Module 3A: Convert all columns to lowercase. Column capitalization sometimes changes between CSV, Google Sheets, and pandas imports. Therefore, we converted variable names to lowercase.

```

[3]: # Module 3A: Convert all column names to lowercase
college_statements.columns = college_statements.columns.str.lower()

# Check the first few rows to ensure the column names are updated
print(college_statements.head())

```

```

      unitid                                instnm \
0  188340      Vaughn College of Aeronautics and Technology
1  188650                                Allen School-Brooklyn
2  188678      American Academy of Dramatic Arts-New York
3  188687  American Academy McAllister Institute of Funer...
4  188890                                Apex Technical School

```

	city	stabbr	zip	latitude	longitud	\
0	Flushing	NY	11369	40.768348	-73.881721	
1	Brooklyn	NY	11201	40.693797	-73.991546	
2	New York	NY	10016	40.745545	-73.984843	
3	New York	NY	10036	40.757105	-73.986466	
4	Long Island City	NY	11101	40.748807	-73.949306	

	mission	\
0	To provide a dynamic learning environment build...	
1	To provide superior, high-quality, campus-base...	
2	A world-leading acting conservatory and colleg...	
3	AAMI exists to: Provide a learning environment...	
4	Offer quality hands-on training that provides ...	

	vision
0	To change the world one student at a time with...
1	Allen College is a Leading Institution to Enri...
2	The American Academy of Dramatic Arts upholds ...
3	None
4	None

Module 3B: Normalize names with Lemmatization. An issue that we may run into is singular and plural words treated as separate words. We normalize the words through the process of lemmatization so that words like student / students, community / communities, program / programs, etc., are treated as the same word before computing TF-IDF.

This keeps the matrix clean and avoids duplicate columns.

```
[4]: # Module 3B: Normalize Text with Lemmatization

from nltk.stem import WordNetLemmatizer
import re

# Initialize lemmatizer
lemmatizer = WordNetLemmatizer()

def normalize_text(text):
    # Convert the text to lowercase and tokenize
    text = text.lower()

    # Use regex to find words (match only alphabetic words)
    words = re.findall(r'\b[a-z]+\b', text)

    # Lemmatize words
    normalized = [lemmatizer.lemmatize(w) for w in words] # Lemmatize the words

    # Return the cleaned and normalized text
```

```

return " ".join(normalized)

# Apply the normalization to the mission statements without renaming the column
college_statements["mission"] = college_statements["mission"].
    ↪ apply(normalize_text)

# Check the first few rows of the cleaned mission text
print(college_statements[["instnm", "mission"]].head())

```

```

                                instnm \
0      Vaughn College of Aeronautics and Technology
1                                Allen School-Brooklyn
2      American Academy of Dramatic Arts-New York
3 American Academy McAllister Institute of Funer...
4                                Apex Technical School

                                mission
0 to provide a dynamic learning environment build...
1 to provide superior high quality campus based ...
2 a world leading acting conservatory and colleg...
3 aami exists to provide a learning environment ...
4 offer quality hand on training that provides y...

```

Module 3C: Describe the Data There are 164 institutions in NYC. There are four CUNY schools that do not have an UNITID assigned per IPEDS

```
[5]: college_statements.describe()
```

```

[5]:      unitid      instnm      city stabbr \
count      162      162      162      162
unique      158      162      21      1
top      Vaughn College of Aeronautics and Technology New York NY
freq      5      1      64      162

      zip latitude longitude \
count      162      162      162
unique      128      156      156
top      11219
freq      5      5      5

                                mission vision
count      162      162
unique      161      33
top      to provide superior high quality campus based ...
freq      2      91

```

Module 3D: Identify which columns have duplicates. Rows 42-45 have duplicated UNITIDs (no entry).

```
[6]: for col in college_statements.columns:
      print(f"\nDuplicates in {col}:")
      print(college_statements[col][college_statements[col].duplicated()])
```

Duplicates in unitid:

```
41
42
43
44
```

Name: unitid, dtype: object

Duplicates in instnm:

Series([], Name: instnm, dtype: object)

Duplicates in city:

```
3          New York
5          Brooklyn
6          New York
7          New York
8    Long Island City
```

...

```
157         New York
158         Brooklyn
159         Brooklyn
160    Forest Hills
161         New York
```

Name: city, Length: 141, dtype: object

Duplicates in stabbr:

```
1      NY
2      NY
3      NY
4      NY
5      NY
```

..

```
157     NY
158     NY
159     NY
160     NY
161     NY
```

Name: stabbr, Length: 161, dtype: object

Duplicates in zip:

```
13    11201
27    10016
29    10065
32    11101
```

35	11201
38	11101
41	10036
42	10027
55	10027
62	10468
67	11101
76	11367
79	11201
88	10027
90	10036
94	10010
101	10036
103	11219
106	11219
109	11211
110	10038
115	11214
116	10019
120	11374
124	10018
126	11219
127	11229
130	11235
135	10001
139	10036
141	10016
145	11230
151	11219
161	10011

Name: zip, dtype: object

Duplicates in latitude:

41	
42	
43	
44	
120	40.72921
136	40.70563

Name: latitude, dtype: object

Duplicates in longitud:

41	
42	
43	
44	
120	-73.858441
136	-74.013948

```
Name: longitud, dtype: object
```

```
Duplicates in mission:
```

```
111    to provide superior high quality campus based ...
```

```
Name: mission, dtype: object
```

```
Duplicates in vision:
```

```
4      None
```

```
5      None
```

```
6      None
```

```
7      None
```

```
8      None
```

```
...
```

```
156
```

```
157
```

```
158
```

```
159
```

```
161
```

```
Name: vision, Length: 129, dtype: object
```

Module 3E: Show Names of Schools that have duplicate UNITIDs. These schools listed below do not have IPEDS UNITID's assigned.

```
[7]: # Schools without UNITIDs

college_statements.loc[college_statements['unitid'].duplicated(keep=False),
↳ ['unitid', 'instnm']]
```

```
[7]:   unitid                                instnm
40              CUNY School of Professional Studies
41              CUNY School of Labor and Urban Studies
42    CUNY Graduate School of Public Health and Heal...
43              CUNY Macaulay Honors College
44    CUNY Craig Newmark Graduate School of Journalism
```

Module 3F: Check the shape, column names, and dtypes of columns

```
[8]: college_statements.shape, college_statements.columns, college_statements.dtypes
```

```
[8]: ((162, 9),
      Index(['unitid', 'instnm', 'city', 'stabbr', 'zip', 'latitude', 'longitud',
            'mission', 'vision'],
            dtype='object'),
      unitid      object
      instnm      object
      city        object
      stabbr      object
      zip         object)
```



```

latitude    object
longitud    object
mission     object
vision      object
dtype: object)

```

Module 3G: Clean the Data Types The variables listed are all objects. We changed them to their appropriate data type.

```

[9]: # Module 3G: Convert Column Data Types

# Convert numeric identifiers
college_statements["unitid"] = pd.to_numeric(college_statements["unitid"],
↳errors="coerce")

# Convert location coordinates
college_statements["latitude"] = pd.to_numeric(college_statements["latitude"],
↳errors="coerce")
college_statements["longitud"] = pd.to_numeric(college_statements["longitud"],
↳errors="coerce")

# Convert text columns to pandas string dtype
text_columns = ["instnm", "city", "stabbr", "zip", "mission", "vision"]

college_statements[text_columns] = college_statements[text_columns].
↳astype("string")

# Verify results
college_statements.dtypes

```

```

[9]: unitid          float64
instnm      string[python]
city        string[python]
stabbr      string[python]
zip         string[python]
latitude    float64
longitud    float64
mission     string[python]
vision      string[python]
dtype: object

```

Module 3H: Convert all string columns to lowercase To avoid misspellings and other problems with mixed lower and upper cases, all string variables are converted to lowercase.

```

[10]: # Module 3H: Convert all string columns to lowercase

```

```
college_statements = college_statements.apply(lambda col: col.str.lower() if
↳col.dtype == "object" else col)
```

```
# Check the first few rows to ensure the conversion
print(college_statements.head())
```

	unitid	instnm \
0	188340.0	Vaughn College of Aeronautics and Technology
1	188650.0	Allen School-Brooklyn
2	188678.0	American Academy of Dramatic Arts-New York
3	188687.0	American Academy McAllister Institute of Funer...
4	188890.0	Apex Technical School

	city	stabbr	zip	latitude	longitud \
0	Flushing	NY	11369	40.768348	-73.881721
1	Brooklyn	NY	11201	40.693797	-73.991546
2	New York	NY	10016	40.745545	-73.984843
3	New York	NY	10036	40.757105	-73.986466
4	Long Island City	NY	11101	40.748807	-73.949306

	mission \
0	to provide a dynamic learning environment buil...
1	to provide superior high quality campus based ...
2	a world leading acting conservatory and colleg...
3	aami exists to provide a learning environment ...
4	offer quality hand on training that provides y...

	vision
0	To change the world one student at a time with...
1	Allen College is a Leading Institution to Enri...
2	The American Academy of Dramatic Arts upholds ...
3	None
4	None

Module 3I: Rename Geocodes for later Mapping To potentially map institutions, the word Longitud is incorrectly spelled and will cause an issue. We rename and spell it correctly.

```
[11]: # Module 3I: Rename goecodes.
```

```
college_statements.rename(columns={"longitud":"longitude"}, inplace=True)
```

Module 4: Save file to a CSV We save the file to a CSV file for later analysis.

```
[12]: # Module 4: Save to CSV
```

```
college_statements.to_csv("college_mission_vision_statements.csv", index=False)
```

```
print("CSV file created successfully.")
```

CSV file created successfully.

Module 5: Prepare the Mission Statement Text Before building a network, we need clean text. This module removes any missing values, and creates a clean text column to be used for analysis.

```
[13]: # Module 5: Prepare Mission Text

import pandas as pd

# Create a clean mission text column
college_statements["mission_clean"] = college_statements["mission"].fillna("").
↳ astype(str)

# View the first few records
college_statements[["instnm", "mission_clean"]].head()
```

```
[13]:                                     instnm \
0      Vaughn College of Aeronautics and Technology
1                                     Allen School-Brooklyn
2      American Academy of Dramatic Arts-New York
3      American Academy McAllister Institute of Funer...
4                                     Apex Technical School

                                     mission_clean
0  to provide a dynamic learning environment buil...
1  to provide superior high quality campus based ...
2  a world leading acting conservatory and colleg...
3  aami exists to provide a learning environment ...
4  offer quality hand on training that provides y...
```

Module 6: Convert Mission Statements into Keywords: Term Frequency - Inverse Document Frequency (TF-IDF) Since text cannot be analyzed directly, we need to convert mission statements into numbers.

Therefore, this module breaks mission statements into individual words.

- We removed common stop words like “the”, “and”, “of”.
- We calculated TF-IDF weights, which measure how important a word is in a document.
- The result is a document-term matrix where rows are institutions and columns are keywords.
- Values are the importance of each word in that mission statement.

TF-IDF reflects how important a word is to a document in a collection of documents (or corpus). It's composed of two parts:

- Term Frequency (TF): This is a measure of the number of times a word appears in a document relative to the total number of words in that document.

- Inverse Document Frequency (IDF): This is a measure of how important a word is within the entire corpus. If a word appears in many documents, it's less significant.

In the output below, one sees mostly zeros because words that don't appear in a given mission statement will have a TF-IDF value of 0. For example, if a particular college's mission statement doesn't include the word "learn", its corresponding entry will be 0 for that word.

Non-zeros are the important words and deemed important for a particular mission statement. For instance, if a mission statement uses the word "academic", it might have a non-zero TF-IDF value, showing that this word is significant for that specific college's mission, relative importance across all mission statements using the IDF component.

```
[14]: # Module 6: Create TF-IDF Matrix

# Import text analysis tools
from sklearn.feature_extraction.text import TfidfVectorizer

# Create vectorizer
vectorizer = TfidfVectorizer(
    stop_words="english",
    min_df=2 # only words that appear in at least 2 documents
)

# Convert mission statements into TF-IDF matrix
tfidf_matrix = vectorizer.fit_transform(college_statements["mission_clean"])

# Extract keyword list
terms = vectorizer.get_feature_names_out()

# Convert to dataframe
tfidf_df = pd.DataFrame(
    tfidf_matrix.toarray(),
    columns=terms
)

tfidf_df.head()
```

```
[14]:
```

	ability	able	academic	academically	academy	accept	access	\
0	0.0	0.0	0.000000	0.0	0.000000	0.0	0.0	
1	0.0	0.0	0.000000	0.0	0.000000	0.0	0.0	
2	0.0	0.0	0.000000	0.0	0.251613	0.0	0.0	
3	0.0	0.0	0.237243	0.0	0.000000	0.0	0.0	
4	0.0	0.0	0.000000	0.0	0.000000	0.0	0.0	

	accessible	accommodate	accomplish	...	world	worthy	written	\
0	0.000000	0.0	0.0	...	0.000000	0.0	0.0	
1	0.000000	0.0	0.0	...	0.000000	0.0	0.0	
2	0.000000	0.0	0.0	...	0.141975	0.0	0.0	

3	0.190874	0.0	0.0	...	0.000000	0.0	0.0
4	0.000000	0.0	0.0	...	0.000000	0.0	0.0

	yaakov	year	yeshiva	yisroel	york	young	youth
0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	0.0	0.0	0.0	0.0	0.0	0.0	0.0

[5 rows x 1103 columns]

Module 7: Build a Two-Mode (Bipartite) Network This module builds a two-node network. We have two types of nodes: Institutions and Mission keywords.

Edges connect institutions to words appearing in their mission statements.

The edges are institution-to-mission keywords.

To build the network, we need to extract the mission keywords that make sense to use in the analysis.

Module 7A: Extracting Keywords

```
[15]: # Module 7A: Extracting Keywords

# Create a DataFrame to store the words with their TF-IDF scores for each
↳ institution
terms = vectorizer.get_feature_names_out()

# Get the TF-IDF scores for each word in each institution's mission statement
tfidf_df = pd.DataFrame(tfidf_matrix.toarray(), columns=terms)

# Extract top keywords for each institution (you can adjust the number of top
↳ keywords to extract)
top_keywords = {}

for index, row in tfidf_df.iterrows():
    top_keywords[college_statements["instnm"].iloc[index]] = row.nlargest(5).
    ↳ index.tolist() # top 5 keywords

# Show the top keywords for each institution
for institution, keywords in top_keywords.items():
    print(f"{institution}: {keywords}")
```

Vaughn College of Aeronautics and Technology: ['inspires', 'built', 'heritage', 'dynamic', 'achieve']

Allen School-Brooklyn: ['assisting', 'continued', 'superior', 'successfully', 'pursue']

American Academy of Dramatic Arts-New York: ['acting', 'actor', 'american',

'conservatory', 'globe']
 American Academy McAllister Institute of Funeral Service: ['profession',
 'entering', 'flexible', 'service', 'academic']
 Apex Technical School: ['marketable', 'graduation', 'hand', 'employment',
 'provides']
 Associated Beth Rivkah Schools: ['jewish', 'associated', 'beth', 'higher',
 'school']
 Bank Street College of Education: ['child', 'education', 'society', 'adult',
 'applying']
 Barnard College: ['advantage', 'new', 'lead', 'woman', 'meet']
 Berk Trade and Business School: ['trade', 'business', 'school', 'theory',
 'student']
 Berkeley College-New York: ['empowers', 'achieve', 'lifelong', 'success',
 'college']
 Beth Hamedrash Shaarei Yosher Institute: ['beth', 'institute', 'torah',
 'talmud', 'study']
 Mount Sinai Phillips School of Nursing: ['nursing', 'delivery', 'sinai',
 'mount', 'equitable']
 Boricua College: ['institution', 'liberal', 'college', 'culture', 'dedicated']
 Brooklyn Law School: ['law', 'legal', 'school', 'dedicated', 'continues']
 St Paul's School of Nursing-Queens: ['nursing', 'providing', 'curriculum',
 'embedded', 'exemplifying']
 Central Yeshiva Tomchei Tmimim Lubavitz: ['chabad', 'central', 'worker', 'life',
 'yeshiva']
 Circle in the Square Theatre School: ['artist', 'actor', 'connects', 'craft',
 'emotional']
 Metropolitan College of New York: ['encourages', 'workplace', 'positive',
 'superior', 'promotes']
 Columbia University in the City of New York: ['operation', 'space', 'campus',
 'facility', 'management']
 The Cooper Union for the Advancement of Science and Art: ['free', 'voice',
 'inspires', 'union', 'discourse']
 Weill Medical College of Cornell University: ['care', 'medicine', 'healthcare',
 'biomedical', 'ground']
 CUNY Bernard M Baruch College: ['driven', 'exploration', 'heart', 'creation',
 'place']
 CUNY Borough of Manhattan Community College: ['community', 'increase', 'inside',
 'borough', 'completion']
 CUNY Bronx Community College: ['choose', 'aspiration', 'bronx', 'citizenship',
 'informed']
 CUNY Brooklyn College: ['art', 'freedom', 'immigrant', 'increasingly',
 'academic']
 CUNY College of Staten Island : ['college', 'higher', 'city', 'mindedness',
 'strength']
 CUNY City College: ['public', 'city', 'transformation', 'visionary', 'york']
 CUNY Graduate School and University Center: ['graduate', 'cuny', 'university',
 'city', 'education']
 CUNY Hostos Community College: ['college', 'community', 'language', 'higher',

'bronx']

CUNY Hunter College: ['creative', 'university', 'seek', 'city', 'research']

CUNY John Jay College of Criminal Justice: ['justice', 'study', 'student', 'prepare', 'global']

CUNY Kingsborough Community College: ['appropriate', 'college', 'centered', 'transfer', 'affordable']

CUNY LaGuardia Community College: ['college', 'urban', 'community', 'learning', 'technology']

CUNY Lehman College: ['social', 'catalyst', 'urban', 'bronx', 'mobility']

CUNY Medgar Evers College: ['culturally', 'producing', 'cuny', 'increasing', 'doe']

CUNY New York City College of Technology: ['tech', 'city', 'technology', 'associate', 'disciplinary']

CUNY Queens College: ['path', 'student', 'prepare', 'aspirational', 'complete']

CUNY Queensborough Community College: ['admission', 'reflective', 'welcoming', 'queen', 'caring']

CUNY School of Law: ['diversity', 'law', 'justice', 'consistently', 'doing']

CUNY York College: ['enables', 'passionate', 'confidence', 'learner', 'engaged']

CUNY School of Professional Studies: ['cuny', 'focusing', 'online', 'study', 'form']

CUNY School of Labor and Urban Studies: ['worker', 'justice', 'public', 'social', 'need']

CUNY Graduate School of Public Health and Health Policy: ['health', 'healthier', 'outcome', 'policy', 'sound']

CUNY Macaulay Honors College: ['curiosity', 'honor', 'inspire', 'transformative', 'lifetime']

CUNY Craig Newmark Graduate School of Journalism: ['ability', 'able', 'academic', 'academically', 'academy']

Yeshiva of Far Rockaway Derech Ayson Rabbinical Seminary: ['moral', 'understanding', 'seek', 'derech', 'considerable']

Fashion Institute of Technology: ['premier', 'adaptable', 'educates', 'sustainability', 'creativity']

Fordham University: ['university', 'transmission', 'catholic', 'wisdom', 'guided']

Helene Fuld College of Nursing: ['nursing', 'single', 'nationally', 'recognized', 'college']

Jewish Theological Seminary of America: ['jewish', 'sharing', 'wisdom', 'order', 'vision']

The Juilliard School: ['fullest', 'music', 'gifted', 'expanding', 'caliber']

LIM College: ['connects', 'partner', 'adaptable', 'relevant', 'close']

Machzikei Hadath Rabbinical College: ['educator', 'rabbinical', 'intent', 'number', 'study']

Mandl School-The College of Allied Health: ['allied', 'health', 'pledge', 'regardless', 'effectively']

Manhattan University: ['university', 'catholic', 'distinctive', 'identity', 'manhattan']

Manhattan School of Music: ['preparing', 'conservatory', 'empowers', 'imaginative', 'music']

Marymount Manhattan College: ['awareness', 'college', 'mission', 'improvement', 'intent']

Memorial Hospital School of Radiation Therapy Technology: ['cancer', 'life', 'ability', 'able', 'academic']

Mesivta Torah Vodaath Rabbinical Seminary: ['men', 'remaining', 'scholastic', 'bible', 'mesivta']

Mesivta of Eastern Parkway-Yeshiva Zichron Meilech: ['teach', 'eastern', 'fold', 'mesivta', 'particularly']

Mesivtha Tifereth Jerusalem of America: ['people', 'guiding', 'involved', 'tifereth', 'aware']

Mirrer Yeshiva Cent Institute: ['highest', 'talmudic', 'scholar', 'teacher', 'ha']

Monroe University: ['national', 'international', 'industry', 'serve', 'align']

University of Mount Saint Vincent: ['physician', 'healthcare', 'assistant', 'culturally', 'mentor']

Icahn School of Medicine at Mount Sinai: ['medicine', 'care', 'health', 'belonging', 'delivery']

The New School: ['understand', 'freedom', 'generating', 'rapidly', 'enables']

New York Law School: ['legal', 'serve', 'various', 'value', 'economy']

New York School for Medical and Dental Assistants: ['aid', 'financial', 'employment', 'clearly', 'assistant']

New York University: ['international', 'outstanding', 'perspective', 'faculty', 'involves']

New York College of Podiatric Medicine: ['medical', 'educational', 'provide', 'medicine', 'clinical']

New York School of Interior Design: ['space', 'design', 'year', 'beautiful', 'dream']

Pace University: ['realize', 'potential', 'university', 'contribution', 'uniquely']

Plaza College: ['promotes', 'advancement', 'dynamic', 'inclusive', 'success']

Pratt Institute-Main: ['advising', 'success', 'undergraduate', 'goal', 'student']

Rabbinical Academy Mesivta Rabbi Chaim Berlin: ['moral', 'understanding', 'seek', 'chaim', 'mesivta']

Rabbinical College Bobover Yeshiva Bnei Zion: ['foster', 'yeshiva', 'analysis', 'text', 'student']

Rabbinical Seminary of America: ['fold', 'america', 'equip', 'position', 'young']

The Rockefeller University: ['benefit', 'biomedical', 'scientist', 'world', 'humanity']

Empire Beauty School-Manhattan: ['beauty', 'empire', 'providing', 'education', 'beginning']

St. Francis College: ['leadership', 'initiative', 'college', 'st', 'expand']

St. Joseph's University-New York: ['st', 'university', 'value', 'tradition', 'rigor']

St Paul's School of Nursing-Staten Island: ['nursing', 'paul', 'career', 'school', 'st']

St. John's University-New York: ['john', 'st', 'university', 'metropolitan',

'global']

SUNY College of Optometry: ['patient', 'care', 'research', 'leadership', 'health']

SUNY Downstate Health Sciences University: ['patient', 'care', 'research', 'leadership', 'health']

SUNY Maritime College: ['adaptive', 'forge', 'evolving', 'thinking', 'critical']

Swedish Institute a College of Health Sciences: ['ll', 'gain', 'needed', 'difference', 'help']

Talmudical Seminary Oholei Torah: ['seminary', 'thoroughly', 'advanced', 'scholar', 'chabad']

Teachers College at Columbia University: ['healthier', 'equitable', 'policy', 'creation', 'engagement']

Torah Temimah Talmudical Seminary: ['ha', 'seminary', 'accommodate', 'reflecting', 'talmudical']

Touro University: ['jewish', 'university', 'promote', 'value', 'component']

Union Theological Seminary in the City of New York: ['union', 'faith', 'justice', 'christian', 'church']

United Talmudical Seminary: ['religious', 'communal', 'lifestyle', 'including', 'educator']

United States Merchant Marine Academy: ['academy', 'federal', 'service', 'state', 'experience']

School of Visual Arts : ['pursuit', 'achievement', 'promotes', 'change', 'citizen']

Wagner College: ['center', 'convenient', 'researcher', 'confident', 'location']

Yeshiva Karlin Stolin: ['moral', 'understanding', 'seek', 'considerable', 'definition']

Yeshiva Derech Chaim: ['chaim', 'derech', 'great', 'yeshiva', 'educational']

Yeshivath Shaar Hatorah: ['including', 'excel', 'hatorah', 'enrollment', 'profit']

Yeshiva University: ['student', 'applying', 'impactful', 'talent', 'start']

Hebrew Union College-Jewish Institute of Religion: ['jewish', 'judaism', 'visionary', 'academic', 'embody']

HJ International Graduate School for Peace and Public Leadership: ['religion', 'wider', 'aspire', 'faith', 'men']

New York Academy of Art: ['academy', 'art', 'artist', 'believe', 'able']

Yeshiva Gedolah Imrei Yosef D'spinka: ['study', 'yeshiva', 'judaic', 'talmudic', 'talmud']

Empire Beauty School-Brooklyn: ['empire', 'beauty', 'student', 'assessment', 'provide']

Albert Einstein College of Medicine: ['dr', 'medical', 'founding', 'school', 'biomedical']

Seminar L'moros Bais Yaakov: ['jewish', 'woman', 'administrative', 'school', 'position']

School of Professional Horticulture, New York Botanical Garden: ['program', 'school', 'training', 'learn', 'practical']

Yeshivas Novominsk: ['yeshiva', 'religious', 'ha', 'dynasty', 'eastern']

Rabbinical College of Ohr Shimon Yisroel: ['higher', 'advocacy', 'organization', 'recognition', 'serf']

Pacific College of Health and Science: ['health', 'medicine', 'science', 'traditional', 'college']

Allen School-Jamaica: ['assisting', 'continued', 'superior', 'successfully', 'pursue']

Charles Stuart School of Diamond Setting: ['dedicated', 'faculty', 'providing', 'director', 'update']

Manhattan School of Computer Technology: ['quality', 'support', 'achievement', 'student', 'need']

New York Conservatory for Dramatic Arts: ['ability', 'able', 'academic', 'academically', 'academy']

Merkaz Bnos-Business School: ['bnos', 'identity', 'religious', 'business', 'campus']

Lia Schorr Institute of Cosmetic Skin Care Training: ['institute', 'care', 'committed', 'craft', 'demanding']

Studio Jewelers: ['licensed', 'easy', 'flexible', 'qualify', 'scheduling']

Yeshiva of the Telshe Alumni: ['torah', 'authentic', 'focusing', 'follower', 'ny']

New York Automotive and Diesel Institute: ['student', 'academic', 'information', 'technical', 'skill']

Metropolitan Learning Institute: ['hand', 'business', 'health', 'support', 'instruction']

Beis Medrash Heichal Dovid: ['medrash', 'lecture', 'study', 'high', 'character']

Aveda Arts & Sciences Institute-New York: ['beauty', 'institute', 'platform', 'transforming', 'known']

Lincoln Technical Institute-Whitestone: ['promise', 'candidate', 'career', 'employer', 'succeed']

Ace Institute of Technology: ['technology', 'creating', 'extensive', 'solution', 'talented']

New Age Training: ['age', 'training', 'effective', 'new', 'individual']

Talmudical Seminary of Bobov: ['chassidic', 'bobov', 'dynasty', 'teaching', 'talmudical']

Center for Allied Health Education: ['healthcare', 'profession', 'allied', 'professional', 'health']

Career School of NY: ['meeting', 'ny', 'special', 'accomplish', 'assist']

Yeshiva of Machzikai Hadas: ['subject', 'study', 'ideal', 'halacha', 'intensive']

EDP School: ['chosen', 'belief', 'student', 'prepare', 'staff']

New York Medical Career Training Center: ['medical', 'dr', 'training', 'field', 'industry']

Institute of Culinary Education: ['hospitality', 'giving', 'today', 'commitment', 'chairman']

Paul Mitchell the School-NYC: ['people', 'board', 'bringing', 'celebrating', 'follow']

Mildred Elley-New York Campus: ['technical', 'skill', 'career', 'reach', 'training']

Atelier Esthetique Institute of Esthetics: ['esthetician', 'skin', 'thriving', 'equips', 'licensure']

Relay Graduate School of Education: ['educator', 'outstanding', 'vision',

'clear', 'team']
 CUNY Stella and Charles Guttman Community College: ['ensure', 'economy',
 'relevant', 'inclusive', 'community']
 Miami Ad School-New York: ['director', 'entering', 'll', 'matter', 'phase']
 Digital Film Academy: ['medium', 'membership', 'like', 'small', 'supportive']
 Sotheby's Institute of Art-NY: ['international', 'industry', 'art', 'expert',
 'facilitating']
 DeVry College of New York: ['shaped', 'talent', 'meaningfully', 'relevant',
 'workplace']
 Rabbinical College Ohr Yisroel: ['ohr', 'yisroel', 'student', 'analysis',
 'purpose']
 Bet Medrash Gadol Ateret Torah: ['sephardic', 'torah', 'center', 'high',
 'custom']
 Yeshiva Ohr Yisrael: ['moral', 'understanding', 'seek', 'ohr', 'considerable']
 Yeshiva Sholom Shachna: ['study', 'talmudic', 'application', 'research',
 'convey']
 Elmezzi Graduate School of Molecular Medicine: ['physician', 'research',
 'scientist', 'conduct', 'provided']
 AMG School of Nursing: ['computing', 'learned', 'unmatched', 'solution',
 'vocational']
 Yeshiva Zichron Aryeh: ['basis', 'yeshiva', 'analysis', 'student', 'method']
 Central Yeshiva Beth Joseph: ['scholar', 'torah', 'important', 'student',
 'appropriate']
 Elyon College: ['individual', 'jewish', 'role', 'institute', 'self']
 Yeshiva Kollel Tifereth Elizer: ['chassidic', 'law', 'action', 'lifestyle',
 'depth']
 Bnos Zion Of Bobov Seminary: ['torah', 'ethical', 'bnos', 'bobov', 'chassidus']
 Transitions Career Institute School of Nursing: ['institute', 'need', 'meeting',
 'increasing', 'transition']
 Congregation Talmidei Mesivta Tiferes Shmiel Aleksander: ['derech', 'yeshiva',
 'chassidische', 'hashem', 'iyun']
 Yeshivat Hechal Shemuel: ['sephardic', 'talmud', 'torah', 'talmudic', 'mussar']
 New York Seminary: ['contemplative', 'imbue', 'productive', 'family', 'sense']
 IE New York College: ['experienced', 'imagine', 'knowledgeable', 'context',
 'educates']
 Yeshiva Yesoda Hatorah Vetz Chaim: ['rabbinical', 'seminary', 'hatorah',
 'yeshiva', 'heritage']
 Yeshiva of Kasho: ['today', 'chassidische', 'life', 'powerful', 'learning']
 Austin Medical Assistant Training: ['health', 'care', 'candidate', 'endeavor',
 'striving']
 Atlantic Acting School: ['acting', 'celebrating', 'company', 'driven',
 'exploring']

Module 7B: Sum the TF-IDF values across all institutions for each term This module extracts the top ten keywords. Note that students and students appear. We will combine the two terms in the next module.

```
[16]: # Module 7B: Sum the TF-IDF values across all institutions for each term
tfidf_sum = tfidf_df.sum(axis=0)

# Sort terms by the total TF-IDF score (highest to lowest)
top_keywords_all = tfidf_sum.nlargest(10)

# Display the top 10 keywords
print("Top 10 keywords across all mission statements:")
print(top_keywords_all)
```

Top 10 keywords across all mission statements:

student	12.820383
education	7.744916
community	7.288431
college	6.187569
learning	5.698280
mission	5.472895
program	5.265152
school	5.216951
provide	5.213416
professional	4.983559

dtype: float64

Module 7C: Exclude specific words This module is designed to exclude any specific words that are deemed superfluous key words. For this analyses we will not exclude any. The code is kept in for further research.

```
[17]: # # Module 7C: Exclude specific words (e.g., "students", "student", "school")
# exclude_words = ["students", "student", "school"]

# # Filter out the terms before summing TF-IDF scores
# filtered_terms = [term for term in tfidf_df.columns if term not in
# ↪exclude_words]

# # Sum the TF-IDF values across all institutions for each filtered term
# tfidf_sum_filtered = tfidf_df[filtered_terms].sum(axis=0)

# # Sort the filtered terms by the total TF-IDF score (highest to lowest)
# top_keywords_filtered = tfidf_sum_filtered.nlargest(10)

# # Display the top 10 filtered keywords
# print("Top 10 keywords across all mission statements (excluding 'students',
# ↪'student', and 'school'):")
# print(top_keywords_filtered)
```

Module 8: Build Bipartite Network with Filtered Keywords (if needed) This module builds a bipartite network of institutions-mission keywords.

```
[18]: # Module 8: Build Bipartite Network with Optional Filtered Keywords

import networkx as nx

# Set this flag to True if you want to filter out specific words (e.g.,
↳ "students", "student", "school")
filter_keywords = False # Set this to True to apply the filter

# List of words to exclude (optional)
exclude_words = ["students", "student", "school"]

# If filtering is enabled, apply the filter
if filter_keywords:
    # Filter out the terms before summing TF-IDF scores
    filtered_terms = [term for term in tfidf_df.columns if term not in
↳ exclude_words]
else:
    # If no filtering is needed, use all terms
    filtered_terms = tfidf_df.columns

# Create empty graph
B = nx.Graph()

# Add institution nodes
for inst in college_statements["instnm"]:
    B.add_node(inst, bipartite="institution")

# Add keyword nodes (filtered if required)
for term in filtered_terms:
    B.add_node(term, bipartite="keyword")

# Add edges between institutions and filtered keywords
for i, inst in enumerate(college_statements["instnm"]):
    for term in filtered_terms:
        weight = tfidf_df.loc[i, term]
        if weight > 0:
            B.add_edge(inst, term, weight=weight)

print("Total nodes:", B.number_of_nodes())
print("Total edges:", B.number_of_edges())
```

Total nodes: 1265

Total edges: 7279

Module 9: Apply the Island Method (Network Reduction) What this module does (plain language)

Large networks are difficult to interpret.

The Island Method works like this:

Imagine the network is a landscape and edge weights are elevations.

As we raise the water level, weaker connections disappear. Only the strongest relationships remain, forming smaller clusters called islands.

This helps reveal core structures in the network.

```
[19]: # Module 9: Edge Threshold Function (Modified)

# Create function to remove weak edges
def trim_edges(G, threshold):
    """
    Remove edges with weight below the threshold to reduce the network to only
    stronger connections.

    Parameters:
    - G (networkx.Graph): The original bipartite graph.
    - threshold (float): The minimum weight to keep an edge.

    Returns:
    - H (networkx.Graph): The graph after applying the threshold.
    """

    # Create a new graph to hold the stronger edges
    H = nx.Graph()

    # Add nodes to the new graph (keeping the original node set)
    H.add_nodes_from(G.nodes(data=True))

    # Iterate through edges and add only those that meet the threshold
    for u, v, data in G.edges(data=True):
        if data["weight"] > threshold:
            H.add_edge(u, v, weight=data["weight"])

    # Optionally, remove isolated nodes (nodes with no edges after trimming)
    isolated_nodes = list(nx.isolates(H))
    H.remove_nodes_from(isolated_nodes)

    return H
```

Module 10: Generate Island Networks This module raises the water level gradually. For each threshold level:

- Weak edges are removed.
- The remaining network splits into smaller islands.

Each island represents a group of institutions sharing strong mission language.

```
[20]: # Module 10: Generate Islands (Modified)

import numpy as np

# Collect edge weights
weights = [d["weight"] for _, _, d in B.edges(data=True)]

# Calculate the minimum and maximum weight for thresholds
mn = min(weights)
mx = max(weights)

# Create threshold levels (you can adjust the number of thresholds)
thresholds = np.linspace(mn, mx, 5)

# List to hold the islands (groups of connected components)
islands = []

# Iterate over each threshold and apply the island method
for t in thresholds:

    # Create the reduced graph by trimming edges based on the threshold
    Gt = trim_edges(B, t)

    # If the graph is not empty, count connected components (islands)
    if len(Gt.nodes()) > 0:
        components = list(nx.connected_components(Gt))

        islands.append((t, Gt, components))

        # Print the current threshold and the number of islands found
        print(f"Threshold: {round(t, 3)}, Nodes: {len(Gt.nodes())}, Islands: ␣
↪{len(components)}")

    else:
        # Skip empty graphs (no nodes after trimming)
        print(f"Threshold: {round(t, 3)} resulted in an empty graph, skipping.")
```

```
Threshold: 0.016, Nodes: 1263, Islands: 1
Threshold: 0.238, Nodes: 509, Islands: 18
Threshold: 0.46, Nodes: 58, Islands: 26
Threshold: 0.682, Nodes: 2, Islands: 1
Threshold: 0.904 resulted in an empty graph, skipping.
```

Module 11: Interpret the Islands What this module does (plain language)

Now we inspect the clusters.

Each island contains:

Institutions

Mission keywords

By looking at the keywords, we can understand the strategic identity of the institutions in that cluster.

```
[21]: # Module 11: Examine Islands (Modified)

# Select the second-to-last island from the islands list (you can adjust based
↳ on your preference)
threshold, Gt, comps = islands[-2] # Adjust index as necessary

# Store results for future analysis (optional)
island_info = []

for i, comp in enumerate(comps):

    # Check if the component is empty
    if len(comp) > 0:
        institutions = [n for n in comp if n in college_statements["instnm"].
↳ values]
        keywords = [n for n in comp if n not in institutions]

        # Store island information
        island_info.append({
            "Island": i + 1,
            "Institutions": institutions,
            "Mission Keywords": keywords
        })

        # Print the results
        print(f"\nIsland {i+1}")
        print("Institutions:")
        print(institutions)
        print("Mission Keywords:")
        print(keywords)
    else:
        print(f"\nIsland {i+1} is empty.")

# Optionally, return the results for further analysis
print("\nIsland Information Stored:")
for info in island_info:
    print(info)
```

Island 1

Institutions:

['Apex Technical School']

Mission Keywords:

['marketable']

Island 2

Institutions:

['Berkeley College-New York']

Mission Keywords:

['empowers']

Island 3

Institutions:

['Beth Hamedrash Shaarei Yosher Institute']

Mission Keywords:

['beth']

Island 4

Institutions:

["St Paul's School of Nursing-Staten Island", 'Mount Sinai Phillips School of Nursing', 'Helene Fuld College of Nursing', "St Paul's School of Nursing-Queens"]

Mission Keywords:

['nursing']

Island 5

Institutions:

['Brooklyn Law School']

Mission Keywords:

['law']

Island 6

Institutions:

['Columbia University in the City of New York']

Mission Keywords:

['operation']

Island 7

Institutions:

['CUNY School of Professional Studies']

Mission Keywords:

['cuny']

Island 8

Institutions:

["Seminar L'moros Bais Yaakov", 'Jewish Theological Seminary of America']

Mission Keywords:

['jewish']

Island 9

Institutions:

['Machzikei Hadath Rabbinical College']

Mission Keywords:

['educator']

Island 10

Institutions:

['Manhattan University']

Mission Keywords:

['university']

Island 11

Institutions:

['Memorial Hospital School of Radiation Therapy Technology']

Mission Keywords:

['cancer']

Island 12

Institutions:

['Mesivta of Eastern Parkway-Yeshiva Zichron Meilech']

Mission Keywords:

['teach']

Island 13

Institutions:

['Empire Beauty School-Manhattan', 'Aveda Arts & Sciences Institute-New York']

Mission Keywords:

['beauty']

Island 14

Institutions:

["St. Joseph's University-New York"]

Mission Keywords:

['st']

Island 15

Institutions:

['SUNY Maritime College']

Mission Keywords:

['forge', 'adaptive']

Island 16

Institutions:

['Swedish Institute a College of Health Sciences']

Mission Keywords:

['ll']

Island 17

Institutions:
['Union Theological Seminary in the City of New York']
Mission Keywords:
['union']

Island 18
Institutions:
['United Talmudical Seminary']
Mission Keywords:
['religious']

Island 19
Institutions:
['United States Merchant Marine Academy']
Mission Keywords:
['academy']

Island 20
Institutions:
['Rabbinical College of Ohr Shimon Yisroel']
Mission Keywords:
['higher']

Island 21
Institutions:
['Ace Institute of Technology']
Mission Keywords:
['technology']

Island 22
Institutions:
['Talmudical Seminary of Bobov']
Mission Keywords:
['chassidic']

Island 23
Institutions:
['Center for Allied Health Education']
Mission Keywords:
['healthcare']

Island 24
Institutions:
['New York Medical Career Training Center']
Mission Keywords:
['medical']

Island 25

Institutions:

['Congregation Talmidei Mesivta Tiferes Shmiel Aleksander']

Mission Keywords:

['derech']

Island 26

Institutions:

['Yeshiva Yesoda Hatorah Vetz Chaim']

Mission Keywords:

['rabbinical']

Island Information Stored:

{'Island': 1, 'Institutions': ['Apex Technical School'], 'Mission Keywords': ['marketable']}

{'Island': 2, 'Institutions': ['Berkeley College-New York'], 'Mission Keywords': ['empowers']}

{'Island': 3, 'Institutions': ['Beth Hamedrash Shaarei Yosher Institute'], 'Mission Keywords': ['beth']}

{'Island': 4, 'Institutions': ["St Paul's School of Nursing-Staten Island", 'Mount Sinai Phillips School of Nursing', 'Helene Fuld College of Nursing', "St Paul's School of Nursing-Queens"], 'Mission Keywords': ['nursing']}

{'Island': 5, 'Institutions': ['Brooklyn Law School'], 'Mission Keywords': ['law']}

{'Island': 6, 'Institutions': ['Columbia University in the City of New York'], 'Mission Keywords': ['operation']}

{'Island': 7, 'Institutions': ['CUNY School of Professional Studies'], 'Mission Keywords': ['cuny']}

{'Island': 8, 'Institutions': ["Seminar L'moros Bais Yaakov", 'Jewish Theological Seminary of America'], 'Mission Keywords': ['jewish']}

{'Island': 9, 'Institutions': ['Machzikei Hadath Rabbinical College'], 'Mission Keywords': ['educator']}

{'Island': 10, 'Institutions': ['Manhattan University'], 'Mission Keywords': ['university']}

{'Island': 11, 'Institutions': ['Memorial Hospital School of Radiation Therapy Technology'], 'Mission Keywords': ['cancer']}

{'Island': 12, 'Institutions': ['Mesivta of Eastern Parkway-Yeshiva Zichron Meilech'], 'Mission Keywords': ['teach']}

{'Island': 13, 'Institutions': ['Empire Beauty School-Manhattan', 'Aveda Arts & Sciences Institute-New York'], 'Mission Keywords': ['beauty']}

{'Island': 14, 'Institutions': ["St. Joseph's University-New York"], 'Mission Keywords': ['st']}

{'Island': 15, 'Institutions': ['SUNY Maritime College'], 'Mission Keywords': ['forge', 'adaptive']}

{'Island': 16, 'Institutions': ['Swedish Institute a College of Health Sciences'], 'Mission Keywords': ['ll']}

{'Island': 17, 'Institutions': ['Union Theological Seminary in the City of New York'], 'Mission Keywords': ['union']}

{'Island': 18, 'Institutions': ['United Talmudical Seminary'], 'Mission

```

Keywords': ['religious']]
{'Island': 19, 'Institutions': ['United States Merchant Marine Academy'],
'Mission Keywords': ['academy']}
{'Island': 20, 'Institutions': ['Rabbinical College of Ohr Shimon Yisroel'],
'Mission Keywords': ['higher']}
{'Island': 21, 'Institutions': ['Ace Institute of Technology'], 'Mission
Keywords': ['technology']}
{'Island': 22, 'Institutions': ['Talmudical Seminary of Bobov'], 'Mission
Keywords': ['chassidic']}
{'Island': 23, 'Institutions': ['Center for Allied Health Education'], 'Mission
Keywords': ['healthcare']}
{'Island': 24, 'Institutions': ['New York Medical Career Training Center'],
'Mission Keywords': ['medical']}
{'Island': 25, 'Institutions': ['Congregation Talmidei Mesivta Tiferes Shmiel
Aleksander'], 'Mission Keywords': ['derech']}
{'Island': 26, 'Institutions': ['Yeshiva Yesoda Hatorah Vetz Chaim'], 'Mission
Keywords': ['rabbinical']}

```

Module 12: Visualize the Bipartite Network This module draws the network. This graph depicts a bipartite network that represents relationships between colleges and the key concepts found in their mission statements. It shows how certain colleges are linked to words or ideas that frequently appear in their mission statements.

- Blue circles (colleges) represent individual colleges. Each one is labeled with the name of a college.
- Orange circles (keywords) represent important ideas or concepts that appear in the mission statements of these colleges, such as “healthcare,” “technology,” “education,” and “union.”
- The lines (edges) between the blue and orange circles indicate that a particular college is focused on the corresponding keyword in its mission statement. For example, the line connecting “BCY” (Berkeley College-New York) to “higher” suggests that the term “higher” appears in its mission statement.
- The layout is organized in such a way that similar institutions or concepts appear near each other, making it easier to spot patterns.

This type of network helps visualize how institutions emphasize different ideas in their mission statements. By analyzing this network, we identify clusters or groups of institutions with similar goals or focus areas, like those focused on healthcare, technology, or education.

One might notice that not all institutions are visible. Only certain schools appear in the graph because the visualization is based on the presence and importance of keywords (derived from their mission statements) and the specificity and frequency of those keywords. Schools with mission statements that have a higher frequency or more emphasis on certain terms (like “healthcare,” “technology,” etc.) are represented, while others may not appear due to less emphasis on those keywords or because their mission statements use different phrasing or are more general.

[22]: `# Module 12: Visualize Network with Abbreviated Names for Institutions and Full
↪Names in Legend (Modified)`

```

import matplotlib.pyplot as plt

# Check if the graph is empty
if len(Gt.nodes()) > 0:
    # Increase figure size for better readability
    plt.figure(figsize=(15, 15)) # Adjust width and height to make the plot
    ↪larger

    # Compute layout only for nodes that exist in Gt
    pos = nx.spring_layout(Gt, seed=42, k=0.3, iterations=50) # Increased 'k'
    ↪value for more space between nodes

    # Extract node types ONLY from Gt
    institutions = [n for n, d in Gt.nodes(data=True) if d.get("bipartite") ==
    ↪"institution"]
    keywords = [n for n, d in Gt.nodes(data=True) if d.get("bipartite") ==
    ↪"keyword"]

    # Abbreviate all institution names by taking the first letter of each word
    abbreviated_institutions = {}
    for inst in institutions:
        abbreviation = ''.join([word[0].upper() for word in inst.split() if
    ↪word]) # Take first letter of each word
        abbreviated_institutions[inst] = abbreviation

    # Draw nodes for institutions
    nx.draw_networkx_nodes(Gt, pos, nodelist=institutions,
                           node_color="lightblue", node_size=700,
    ↪label="Institution")

    # Draw nodes for keywords
    nx.draw_networkx_nodes(Gt, pos, nodelist=keywords,
                           node_color="orange", node_size=500, label="Mission
    ↪Keyword")

    # Draw edges
    nx.draw_networkx_edges(Gt, pos, alpha=0.5)

    # Draw abbreviated labels for institutions
    nx.draw_networkx_labels(Gt, pos, labels=abbreviated_institutions,
    ↪font_size=10)

    # Draw full labels for keywords
    keyword_labels = {keyword: keyword for keyword in keywords}

```

```

    nx.draw_networkx_labels(Gt, pos, labels=keyword_labels, font_size=10,
↪font_color="black")

    # Add custom legend with full names for institutions
    custom_labels = [{"abbr": full_name} for full_name, abbr in
↪abbreviated_institutions.items()]

    # Creating custom handles for the legend (one for each institution)
    custom_handles = [plt.Line2D([0], [0], marker='o', color='w', label=label,
↪markersize=10, markerfacecolor='lightblue')
        for label in custom_labels]

    # Display the custom legend on the right side
    plt.legend(custom_handles, custom_labels, loc="upper left",
↪bbox_to_anchor=(1, 1))

    # Title and cleanup
    plt.title("Mission Statement Bipartite Network (Island Method)",
↪fontsize=16)
    plt.axis("off") # Turn off axis for a cleaner look
    plt.show()

else:
    print("The graph is empty. No nodes to visualize.")

```



